Enhancing Open-Domain Question Answering with Hypothetical Document Generation

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Abstract

This paper is devoted to exploring the potential of enhancing open-domain question answering (QA) systems with hypothetical document generation. While Large Language Models (LLMs), such as Llama, have made significant progress in QA, they often struggle with outdated or insufficient knowledge. To address this, synthetic documents were generated using Mistral 7B-Instruct and incorporated into the pipeline. The effectiveness of this approach was evaluated on the TriviaQA validation set by comparing Llama 3.2's performance in three settings: without additional context, with generated documents, and with retrieved Wikipedia passages. Several prompts were tested to reduce over-reliance on provided context and better balance the use of internal and external knowledge. While the usage of hypothetical documents did not improve exact match scores, it led to noticeable gains in BERTScore, indicating semantic relevance. The study highlights the promise of this method, particularly in low-resource or retrieval-limited scenarios, and discusses challenges.

Keywords: *Open-Domain Question Answering*, *Hypothetical Document Generation*, *Retrieval-Augmented Generation*, *Llama 3.2*, *Mistral 7B-Instruct*, *TriviaQA*

1 Introduction

Open-domain question answering has seen remarkable progress in recent years, driven largely by the development of Large Language Models, or LLMs. While models like GPT-4 are capable of understanding and generating human-like responses, they still face limitations when applied to open-domain QA tasks. One key problem here is their substantial reliance on the knowledge from their parameters, which can become outdated or insufficient over time.

The concept of Retrieval-Augmented Generation, or RAG, has emerged in response to these limitations. RAG combines the power of information retrieval with generative models, allowing them to access external and up-to-date knowledge that they lack to generate more accurate and relevant answers.

An alternative to the retrieve-then-read pipeline is the concept of generating hypothetical documents. In this paradigm, instead of relying solely on retrieved content, the model generates plausible documents or contexts that might contain the answer. Creating synthetic data has several benefits: while it reduces reliance on large retrieval corpora, it still enables the generation of context tailored to specific questions and, thus, has potential to improve performance. This can be especially helpful in situations where retrieval or resources are limited. By generating hypothetical documents, the model can simulate a broader range of possible answers improving flexibility and robustness in open-domain QA tasks. This represents a shift in the paradigm, offering an interesting alternative or even a complement to the traditional retrieve-then-read approach.

By generating synthetic data, we can make the most of the knowledge that is already incorporated in the model's parameters. Indeed, LLMs have been trained on vast amounts of data, and this embedded knowledge can be leveraged to generate contextually relevant content. This approach maximizes the utility of the model's parameters as they are, without the need for extensive external resources in situations where resources are limited.

In this work, the following contributions are made:

- **Evaluation of Hypothetical Document Generation for Open-Domain QA** – This work investigates whether generating synthetic documents as additional context improves the performance of Llama 3.2 on TriviaQA. The experiments compare model performance under three conditions: without additional context, with generated documents from Mistral 7B-Instruct, and with Wikipedia passages.
- **Analysis of Model Over-Reliance on Provided Context** – The study highlights how the model may overtrust provided documents, sometimes failing even when they likely know the answer from their internal knowledge.

2 Related Works

This section reviews key research on Retrieval-Augmented Generation, covering its integration with language models, improvements in retrieval, scalability, and the emerging generate-then-read approach.

2.1 Retrieval Augmented Generation

One of the first works to introduce the idea of combining retrieval with language models was REALM Guu et al. (2020). Unlike traditional models that store all the knowledge implicitly in their parameters, REALM allows the model to decide what external knowledge to retrieve and use during inference. The model first retrieves relevant documents from a large corpus, such as Wikipedia, and then processes these documents to form the output. This retrieval step is integrated into the model's training process, requiring backpropagation through the retrieval mechanism and the entire corpus of textual knowledge. Due to this strategy, a model can utilize external knowledge and improve performance on knowledge-intensive tasks.

The concept of Retrieval Augmented Generation (RAG) was then formulated in the Lewis et al. (2020). The approach combines pre-trained language models with external knowledge by using both parametric and non-parametric memory. The architecture includes a seq2seq model, like BART, for generating outputs, and a dense vector index accessed by a retriever (DPR) to find relevant documents. The model is trained end-to-end, with the retriever and generator fine-tuned together. RAG's main goal is to allow models to use external information and, therefore, enhance the model's ability to answer complex questions.

2.2 Retrieve-then-Read pipeline

The key idea behind the retrieve-then-read pipeline is to first focus on gathering the right information (a retrieval component) and then process that information to generate an accurate answer (a reading component).

After the development of RAG, further works focused on improving its components: either the retrieval or the reading one. Since retrieval quality heavily impacts generation performance, the need for better retrieval methods has emerged. For instance, in Karpukhin et al. (2020), the authors focus on improving document retrieval for open-domain question answering by introducing Dense Passage Retrieval (DPR), a neural retrieval method that uses dense vector representations and a dual-encoder architecture to enhance retrieval accuracy and efficiency.

With the rapid development of different RAG implementations, another direction of research has focused on improving scalability. With the RePLUG Shi et al. (2023) framework, a retrieval component became highly accessible even for black-box language models, appearing as a separate plug-and-play module. This allowed the use of RAG without retraining the entire system, reducing computational costs and making such models more practical for real-world applications.

2.3 Generate-then-Read pipeline

The generate-then-read paradigm is an emerging approach in retrieval-augmented systems that replaces traditional document retrieval with synthetic context generation. This is an entirely different perspective of equipping language models with knowledge.

Unlike the previous paradigm, this one leverages the implicit knowledge stored in LLMs themselves. Thus, it reduces dependency on external databases while still providing relevant and informative contexts. Compared to retrieve-then-read approaches, generate-then-read can mitigate issues like retrieval errors and domain mismatch. However, its main advantage lies in low-resource settings, allowing better generation without collecting large corpora. Several challenges in this paradigm are related to ensuring the factuality and diversity of generated contexts.

In the Yu et al. (2023), the authors propose that instead of retrieving external documents, LLMs can generate hypothetical documents tailored to specific questions. In the paper, two main advantages of this approach are highlighted. Firstly, generated documents are more likely to contain the correct answer than retrieved ones, as LLMs use deep token-level attention between a question and a document, making the content more specific to the question. Secondly, generating contextual documents outperforms directly generating answers even without external information. This task aligns with the model's training objective, and therefore allows it to better utilize its stored world knowledge.

Another work that leverages the concept of hypothetical documents generation is Gao et al. (2022). In their HyDe approach, the focus is on generating hypothetical documents to improve retrieval performance in a zero-shot setting. Unlike traditional methods that rely on labeled data, HyDe generates synthetic documents that can serve as relevant context for a given query. The authors highlight two main advantages of this method. First, generating hypothetical documents eliminates the need for manually labeled relevance data, making the approach more scalable and adaptable to new tasks. Second, the approach improves retrieval accuracy by using the generated documents, which are more tailored to the specific query compared to traditional retrieval-based methods.

2.4 Other approaches

Some research is focused on other parts of the pipeline. For instance, in Ma et al. (2023), the authors adapt the search query itself instead of improving either reader or retriever. By prompting another LLM to generate a more nuanced and precise query, they managed to achieve consistent improvements in open-domain QA and downstream tasks.

Some works, such as Asai et al. (2023), introduce completely different approaches. SELF-RAG is a self-reflective framework for retrieval-augmented generation where the model critiques and refines its own answers. It performs initial retrieval and generation, then assesses whether the retrieved context is sufficient. This approach improves factual accuracy, especially when initial retrievals are noisy or incomplete.

Thus, there exist multiple ways and perspectives to improve RAG systems. This work focuses on generate-then-read pipeline, since it does not require creating large corpora for open-domain QA, yet it shows prominent results and is highly accessible.

3 Methods

This section outlines the methodology used to enhance open-domain question answering with hypothetical document generation. Specifically, it details the approach for synthetic document generation, model integration, and evaluation metrics.

3.1 Hypothetical Document Generation

The primary goal of this work is to generate contextually relevant documents that could act as additional input for the question-answering process. This is achieved using the Mistral 7B-Instruct model, which generates synthetic documents based on a given question. Mistral 7B-Instruct is a large-scale language model fine-tuned for instruction-following tasks. It has the ability to create coherent and contextually rich content, making it suitable for generating hypothetical documents.

Thus, the prompt to Mistral 7B-Instruct is formulated in such a way that it guides the model to generate a relevant document that may contain the answer.

3.2 Model Description

The Llama 3.2 model is used as the base model for answering questions in this study. Llama 3.2 is a state-of-the-art open-domain question-answering model with 3 billion parameters. It has been pre-trained on a vast corpus of text and fine-tuned for QA tasks. Llama 3.2 excels at generating human-like answers, but its performance can be limited by the quality of the context it sees during inference.

In this work, hypothetical documents generated by Mistral 7B-Instruct are integrated into the Llama 3.2 model's inference pipeline. Specifically, the model is given the original question along with the generated document as additional context, which is expected to enhance the model's ability to answer the question accurately.

3.3 Wikipedia Passages

In order to explore the effect of equipping Llama with additional documents more thoroughly, corresponding Wikipedia passages were given along with the question. Wikipedia passages were extracted in the following way: first, the question was searched through the Wikipedia part of TriviaQA. If it existed, its Wikipedia document was passed as an additional context to the model. If there was no passage for a particular question in the dataset, a short summary from Wikipedia related to a given question was retrieved using Wikipedia API. As mentioned in Yu et al. (2023), generated documents are more likely to contain relevant information than retrieved ones. In this work, Wikipedia was incorporated in order to check whether this statement is true.

Although the documents are not chosen dynamically and are assumed to already contain relevant context needed to answer a question, it still provides useful insights about how the model perform with different kinds of supportive contexts.

3.4 Evaluation Setup

In terms of data, TriviaQA was chosen for the experiments as it offers a diverse set of factual questions with associated web passages. The dataset contains over 650k question-answer pairs across various topics, making it suitable for evaluating the model. However, due to resource constraints, all the models were evaluated on the same set of 1000 samples.

The performance of Llama 3.2 is evaluated under three conditions:

1. Baseline (without additional context)
2. Llama 3.2 with generated documents from Mistral 7B-Instruct
3. Llama 3.2 with Wikipedia passages

The following metrics are used to assess the model's performance: Exact Match (EM) and BERTScore Zhang et al. (2020).

The Exact Match score is a classical evaluation metric in open-domain QA tasks. It measures the percentage of questions for which the model's answer exactly matches the ground truth answer. While EM provides a strict evaluation, it does not capture semantically correct answers that may differ in wording but convey the same information. Therefore, the second metric was used to complement EM.

BERTScore is a more flexible metric that evaluates the semantic similarity between predicted and reference answers using contextual embeddings from a pre-trained BERT model. Unlike EM, BERTScore captures semantic meaning, making it suitable for situations where different wording may still convey the same answer. The default model used for BERTScore is "roberta-large".

4 Experiments and Results

To obtain documents from Mistral-7B-Instruct, the following prompt was used: *"Generate a detailed passage to help answer this question: {question}"*.

Below are a couple of examples of generated documents along with the respective questions:

Q: What was the name of Michael Jackson's autobiography written in 1988?
A: Moonwalk

Michael Jackson's autobiography, written in 1988, was titled "Moonwalk." The book was a detailed account of Jackson's life and career, from his early days as a child performer to his rise to fame as the "King of Pop." In "Moonwalk," Jackson opened up about his personal life, including his struggles with depression and his relationship with his family. He also discussed his creative process and his love for music. The book was a critical and commercial success, spending 26 weeks atop the New York Times Best Seller list. It remains one of the best-selling autobiographies of all time.

Q: What star sign is Jamie Lee Curtis?
A: Scorpio (born on November 22nd)

Jamie Lee Curtis is a well-known actress and producer, born on December 22nd, 1958. She hails from Los Angeles, California, and has been active in the entertainment industry for over four decades. Curtis is best known for her roles in horror films such as Halloween, The Fog, and Tremors, but she has also appeared in a variety of other genres, including comedy, drama, and action. Curtis is a Sagittarius, which is the ninth sign of the zodiac. Sagittarians are known for their adventurous and optimistic nature, as well as their love of learning and exploration. They are often seen as philosophical and free-spirited, and they enjoy traveling and seeking out new experiences. In addition to her acting career, Curtis is also an accomplished author, having written several children's books, including the popular series "The Very Hungry Caterpillar." She is also a devoted advocate for environmental causes and has worked with various organizations to promote sustainability and conservation. Overall, Jamie Lee Curtis is a Sagittarius, born on December 22nd, 1958. She is known for her adventurous spirit, love of learning, and commitment to environmental causes.

As can be observed from the second example, Mistral-7B-Instruct tended to hallucinate. Besides, it sometimes generated short or irrelevant answers instead of full documents. For instance:

Q: What is the only novel to top the best-seller lists for two consecutive years?
A: Jonathan Livingston Seagull

Fifty Shades of Grey by E.L. James

Q: Which female singer was born on exactly the same day as impressionist Rich Little?
A: Tina Turner

There is no female singer who was born on exactly the same day as Rich Little, as Rich Little was born on July 21, 1942, and there are no notable female singers who were born on that exact same day.

Once the documents were obtained, the model was tested in all three conditions mentioned in 3.4. The baseline was tested using the following prompt: *"You are given a question. Output only the final answer, do not explain your reasoning or include any additional text. The question is: {question}".*

For evaluating Llama 3.2 in other two settings, three different prompts were used. The prompts are listed below. All of them, except the last one, are zero-shot.

Prompt 1

"You are given a document that provides useful context for answering the question. Answer the following question and output only the final answer. Do not explain your reasoning or include any additional text.

The document is: {document}.
The question is: {question}"

Prompt 2

"You are given a question and a document that may help answer it. Use the document if it contains relevant information. If not, answer using your own knowledge. Output only the final answer, do not explain your reasoning or include any additional text.
The document is: {document}.
The question is: {question}.

Final Answer:"

Prompt 3

"You are given a question and a document. The document may be helpful, but it may not contain all the information needed. Use the document if it contains relevant information. If not, answer using your own knowledge. If you do not know, just provide the best possible answer. Output only the final answer as concisely as possible, do not include any explanations, reasoning or extra text.

Example:
Question: Which American-born Sinclair won the Nobel Prize for Literature in 1930?
Answer: Harry Sinclair Lewis.

The document is: {document}.
The question is: {question}.

Final Answer:"

The prompts and the reason why they were designed this way are discussed in the upcoming section 5. All values of the evaluation metrics can be found in Table 1.

Model	EM	BERTScore
Baseline		
Llama 3.2-3B	**0.32**	0.7522
Prompt 1		
Llama 3.2-3B + Mistral-7B-Instruct	0.25	0.7798
Llama 3.2-3B + Wikipedia	0.18	0.7352
Prompt 2		
Llama 3.2-3B + Mistral-7B-Instruct	0.26	0.7798
Llama 3.2-3B + Wikipedia	0.23	0.7522
Prompt 3		
Llama 3.2-3B + Mistral-7B-Instruct	0.25	**0.7798**
Llama 3.2-3B + Wikipedia	0.21	0.7522

Table 1: Performance comparison of different prompts

5 Discussion

While exploring different types of prompts and settings, several interesting observations were made. Firstly, the model (Llama 3.2) relied too much on the provided documents. If an answer was not clearly stated in the text, the model often failed to produce a correct response when it likely knew the answer from its internal knowledge. This suggested that the way prompts are phrased plays a crucial role in how much the model relies on the provided context.

The first prompt told the model that a document was useful and should contain an answer to a question. Although this generally makes sense, in practice, it led the model to stick too closely to the document, even when it was not helpful. Thus, there were many responses like *"There is not enough information provided in the document to accurately answer this question"*.

The second prompt was a bit more flexible, as it told the model to use the document if it contained relevant information, otherwise to rely on its own knowledge. This led to slight improvements, especially when using Wikipedia as the document source.

The third prompt added even more guidance. It made it clear that the document might not have all the information and included an example to show the model exactly what kind of answer was expected. Besides, the goal was to encourage the model to output any factual answer even if it was not entirely sure, to avoid answers like "I do not know" as much as possible. While this prompt slightly reduced the EM scores for both models, it led to the best value of BERTScore, although it may not be obvious from the table.

Exact Match is commonly used in open-domain question answering for several reasons, but it has limitations as a metric. Since EM relies on exact word matches between the predicted answer and the ground truth, even minor differences can result in a score of 0. This strict evaluation can be unfair, especially when a model's answer is semantically correct. To address this issue, BERTScore was also used, providing a more flexible evaluation by comparing the semantic similarity of the answers. One of the examples from the dataset, where it was especially difficult for the model to output the ground truth answer:

Q: How many different animal shapes are there in the ""Animal Crackers"" cookie zoo?

Answer (dataset, not preprocessed): Eighteen–two bears (one walking, one seated), a bison, camel, cougar, elephant, giraffe, gorilla, hippopotamus, hyena , kangaroo, lion, monkey, rhinoceros, seal, sheep, tier, and zebra

Answer (Llama 3.2 + Mistral documents): the document doesnt provide an exact number of different animal shapes however it mentions a variety of popular animals such as giraffes zebras monkeys bears and tigers as well as some lesserknown animals like kangaroos and koalas it suggests that there are countless animal shapes in the cookie zoo implying an almost endless diversity of options

Answer (Llama 3.2 + Wiki): there is no information about an animal crackers cookie zoo so we cannot determine the number of different animal shapes

Here, even if the model had guessed the answer right and outputted "18" or "eighteen", it would not have been counted. However, a semantic-based metric like BERTScore would capture this and be more suitable. Another example of a reference answer, which is phrased in such a way that makes it hard for a model to reply correctly:

Q: Complete this Biblical quotation: ""It is easier for a camel to go through the eye of a needle, than...

Answer (dataset): ...for a rich man to enter into the kingdom of God. The words are those of Jesus, from Matthew 19:24

The model managed to continue the quotation, however, the dataset answer included unnecessary clarifications. The question simply asks to **complete** the quote.

Even though the absolute values of EM may not be representative in some cases, several conclusions can be made based on the Table 1. First, synthetic documents, generated by another model-assistant, can achieve an improvement in the clarity of the answers (at least, at a semantic level). The metrics indicate that they can be more helpful than providing the corresponding Wikipedia passages. Although the EM scores declined when using additional documents compared to the baseline value, BERTScore followed the reverse upward trend, slightly improving as the prompts became more refined.

In terms of the prompt designs, the model appeared to be sensitive to the formulations, demonstrating over-reliance on the provided documents. It is interesting how the model becomes incapable of using its own knowledge when it comes to helping it with some additional information. In this paper, only three prompts were used to test that, so further experiments need to be conducted, but it is a very exciting area of research.

6 Limitations

There are a few important limitations to note in this study:

- **Model Limitations**: The criteria for the models in this work were the following: a model should be open-source and free of charge due to the limited resources. While both Llama 3.2 and Mistral 7B-Instruct are powerful models, their ability to generate contextually accurate documents and responses depends heavily on the quality and specificity of the prompts. Instead, more powerful models could be tested, such as InstructGPT.
- **Synthetic Document Accuracy**: Generated documents are not guaranteed to be factually correct, and there is a risk of introducing errors or irrelevant information into the model's context. Reliance on synthetic data for QA tasks can indeed be controversial and risky.

- **Dataset Size**: Due to resource constraints, only 1,000 questions from the TriviaQA validation set were used. A larger evaluation set could potentially yield more robust conclusions.

7 Conclusion

This study explored the use of hypothetical document generation as an alternative to traditional retrieval-based methods in open-domain question answering. By providing Llama 3.2 with synthetic documents generated by another language model, it was possible to evaluate how such context affects performance compared to Wikipedia passages and no additional context. The results showed that while Exact Match scores did not always improve, BERTScore, a more flexible, semantic-based metric, suggested enhanced answer quality when using generated documents.

The experiments also revealed a tendency of the model to rely too heavily on provided context, occasionally overlooking answers it likely knew internally. This over-reliance varied depending on prompt phrasing, emphasizing the importance of instruction design in retrieval-augmented settings.

Overall, the findings support the idea that synthetic documents can serve as a decent supplement, or even an alternative, to retrieved content in open-domain QA. This approach uses the model's own knowledge and opens up flexible opportunities for future research, where prompts, generation quality, and combined retrieval-generation methods can be improved.

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